

Site-specific DNA demethylation during spermatogenesis presets the sites of nucleosome retention in mouse sperm

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eLife Assessment

This study provides **valuable** findings on the role of site-specific DNA methylation changes during spermatogenesis and their contribution to paternal epigenetic inheritance. The study proposes that selective loss of DNA methylation at a subset of promoters is required for nucleosome retention and the establishment of epigenetic states that may influence embryonic gene regulation. The present study's conclusion is mostly supported by **solid** data.

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Abstract

DNA methylation patterns are inherited from the parental germline to the embryo. In mature sperm, the sites of unmethylated DNA are tightly coupled to sites of histone retention at gene regulatory elements that are implicated in paternal epigenetic inheritance. The timing and mechanism of site-specific DNA demethylation in the male germline currently remains unknown. Here, we perform genome-wide profiling of DNA methylation during spermatogenesis by capturing methylated DNA through interaction with a methyl-DNA binding protein domain (MBD). Our data demonstrate that there is a site-specific change in DNA methylation during the mitosis-to-meiosis transition. Importantly, the genomic sites that are demethylated during this transition predetermine nucleosome retention sites in spermatozoa. These results suggest that site-specific DNA demethylation during the mitosis-to-meiosis transition of spermatogenesis prepares embryonic gene expression after fertilization. We therefore propose DNA demethylation during spermatogenesis as a novel phase of epigenetic reprogramming that contributes to embryonic gene regulation.

Introduction

Methylation at the 5-carbon position of cytosine (5-methylcytosine: 5mC) at CpG dinucleotides in DNA is dynamically reprogrammed in the parental germline and contributes to epigenetic inheritance from gametes to embryos (Greenberg & Bourc'his, 2019). In mammalian sperm, the vast majority of histones are replaced by protamines, but sites of unmethylated DNA are tightly coupled to sites of histone retention at gene regulatory elements (Erkek et al., 2013). These gene regulatory elements are associated with the embryonic gene expression program after fertilization (Brykczynska et al., 2010; Carone et al., 2014; Erkek et al., 2013; Hammoud et al., 2009; Jung et al., 2019; Jung et al., 2017; Yamaguchi et al., 2018; Yoshida et al., 2018) and therefore contribute to paternal epigenetic inheritance (Gaspa-Toneu & Peters, 2023; Lismer & Kimmins, 2023). However, it remains unknown when specific sites of DNA hypomethylation are established in the male germline to predetermine histone retention sites in spermatozoa.

The sites of histone retention exhibit features of bivalent genomic domains. These domains are characterized by concomitant enrichment of repressive Polycomb Repressive Complex 2 (PRC2)-mediated trimethylation of histone H3 at lysine 27 (H3K27me3) and active di/trimethylation of H3 at lysine 4 (H3K4me2/3) (Brykczynska et al., 2010; Erkek et al., 2013; Hammoud et al., 2014; Hammoud et al., 2009; Lesch et al., 2013; Sachs et al., 2013). Recent studies have demonstrated that alteration of the H3K27me3 and H3K4me3 levels in sperm leads to changes in gene expression in the next generation (Lesch et al., 2019; Lismer et al., 2021; Sakashita et al., 2023; Siklenka et al., 2015).

In the male germline, DNA methylation is largely reprogrammed in primordial germ cells. It globally increases in embryonic prospermatogonia and reaches a level similar to that in spermatozoa in postnatal spermatogonia (~80% of all CpG sites) (Hammoud et al., 2014; Kobayashi et al., 2013; Kubo et al., 2015; Seisenberger et al., 2012). In prospermatogonia, DNA methylation outside of CpG islands, i.e. clusters of CpG dinucleotides that are constitutively hypomethylated, is established through the action of the histone methyltransferase NSD1, the de novo DNA methyltransferase DNMT3A and its catalytically inactive cofactor DNMT3L (Kaneda et al., 2004; Kato et al., 2007; Shirane et al., 2020). Evolutionarily young transposable elements are recognized by the PIWI-interacting RNA (piRNA) pathway and methylated by the germline-specific de novo DNA methyltransferase DNMT3C (Aravin et al., 2007; Barau et al., 2016; Jain et al., 2017; Kuramochi-Miyagawa et al., 2008).

Based on the studies of paternal genomic imprinting loci, it has long been thought that DNA methylation is stably maintained throughout spermatogenesis following its acquisition in prospermatogonia (Abramowitz & Bartolomei, 2012; Sasaki & Matsui, 2008; Schaefer et al., 2007). Nowadays, mounting evidence suggests a more dynamic nature of DNA methylation during postnatal spermatogenesis. After the postnatal stage, changes in DNA methylation were initially detected in testicular germ cells (Oakes et al., 2007) and during spermatogonial differentiation (Shirakawa et al., 2013). Later studies clarified that, during spermatogenesis, male germ cells undergo a global transient reduction of DNA methylation in early meiotic prophase I (Gaysinskaya et al., 2018; Huang et al., 2023). This process is presumably passive and caused by a delay in establishing maintenance DNA methylation by DNMT1 and its cofactor UHRF1 during premeiotic DNA replication (Gaysinskaya et al., 2018; Huang et al., 2023). Other genome-wide studies have also detected modest changes in DNA methylation during spermatogenesis (Ben Maamar et al., 2022; Chen et al., 2020; Liu et al., 2019; Siebert-Kuss et al., 2024). However, the extent of site-specific regulation of DNA methylation during postnatal spermatogenesis beyond the global transient reduction remains unclear.

To detect site-specific changes in DNA methylation during spermatogenesis, we performed MethylCap-seq during representative stages of spermatogenesis. This approach employs capture of methylated DNA via the Methyl-CpG-binding domain (MBD) followed by next-generation sequencing (Brinkman et al., 2010 [DOI](#)). Our analysis shows that a site-specific loss of DNA methylation during the mitosis-to-meiosis transition predetermines nucleosome retention sites in mature sperm. Based on these results, we propose that meiosis is a process of epigenetic reprogramming that sets up embryonic gene regulation.

Results

DNA methylation dynamics during spermatogenesis

During mouse spermatogenesis, a transient reduction of DNA methylation occurs globally in the premeiotic S phase. The level of DNA methylation is then gradually recovered in meiotic prophase I (Gaysinskaya et al., 2018 [DOI](#); Huang et al., 2023 [DOI](#)) (Fig. 1A [DOI](#)). To evaluate the overall change before and after this transient reduction of DNA methylation, we first quantified the amount of 5mC during spermatogenesis using an enzyme-linked immunosorbent assay (ELISA). Specifically, we analyzed postnatal day 7 (P7) THY1⁺ undifferentiated spermatogonia and KIT⁺ differentiating spermatogonia, both cellular stages prior to the transient reduction. To cover stages of late spermatogenesis after the transient reduction of DNA methylation, we analyzed meiotic pachytene spermatocytes (PS) and postmeiotic rounds spermatids (RS) from adult testes. We found the 5mC level increased during the initial spermatogonial differentiation (Fig. 1B [DOI](#)). After the transient reduction, 5mC levels in PS did not recover to the level of KIT⁺ differentiating spermatogonia, but after meiosis, they slightly increased in RS (Fig. 1B [DOI](#)).

To independently evaluate these observations, we performed immunohistochemistry on testicular sections from adult mice. In accordance with the ELISA results, the nuclear 5mC signals increased in differentiating spermatogonia that are devoid of PLZF, a marker of undifferentiated spermatogonia (Fig. 1C [DOI](#)). These results are consistent with the earlier observation that 5mC levels increase during spermatogonial differentiation (Shirakawa et al., 2013 [DOI](#)). In the meiotic prophase, leptotene/zygotene spermatocytes show bright 5mC staining. In contrast, the 5mC signal is diminished in pachytene/diplotene spermatocytes that express the testis-specific histone variant H1T, a marker that is induced after the mid-pachytene stage (Fig. 1D [DOI](#)). These results suggest that 5mC levels change dynamically during spermatogenesis before and after the transient reduction of DNA methylation in premeiotic S phase.

Detection of site-specific DNA methylation dynamics in spermatogenesis

Given these dynamic genome-wide methylation changes during spermatogonial differentiation and subsequent meiosis, we sought to determine site-specific 5mC changes. We therefore performed MethylCap-seq (Brinkman et al., 2010 [DOI](#)), which, unlike whole-genome bisulfite sequencing (WGBS), which does not distinguish between 5mC and 5-hydroxymethylcytosine (5hmC), specifically detects 5mC. While MethylCap-seq does not offer base-pair resolution, it does provide overall profiles of 5mC genome-wide, particularly on dense CpG areas (Nair et al., 2011 [DOI](#)). In addition, we reasoned that because MethylCap-seq depends on the recognition of 5mC by MBD, the readout of this method represents the recognition by MBD, which may be relevant to functional aspects of 5mC.

We performed MethylCap-seq in wild-type P7 THY1⁺ and KIT⁺ spermatogonia and adult PS and RS with two biological replicates. We detected approximately 130,000 to 200,000 common peaks genome-wide at each transition (Fig. 2A [DOI](#)), demonstrating that the overall MethylCap-seq profile largely overlapped between each stage. This observation is in line with the fact that male germ

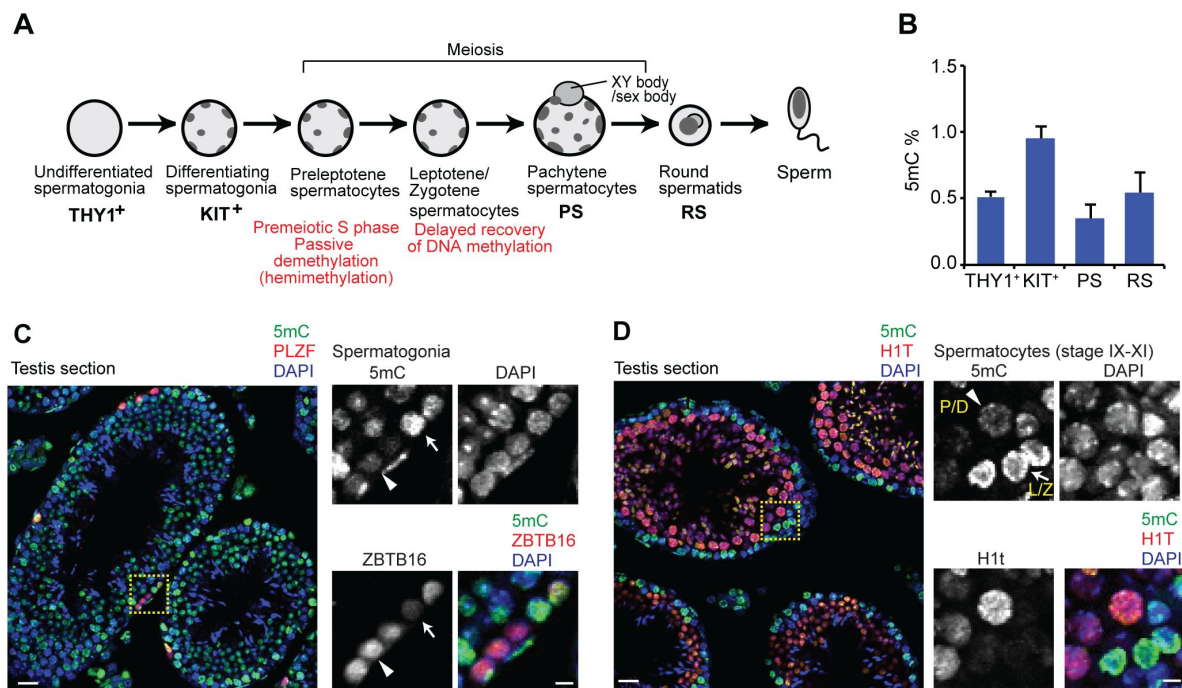


Figure 1

Global characteristics of DNA methylation during spermatogenesis.

(A) Schematic of spermatogenesis and the four representative stages analyzed in this study ($THY1^+$, KIT^+ , PS , and RS). The time window of the transient reduction of DNA methylation is indicated.

(B) ELISA-based quantification of 5-mC during spermatogenesis. Percentage of 5mC in total DNA is shown.

(C, D) Immunostaining of paraffin sections of adult testes with anti-5mC antibody and antibodies against stage-specific markers (ZBTB16 and H1T). Slides were counterstained with DAPI. Images were acquired with a confocal microscope. Regions bordered by dashed yellow squares are magnified in the right panels. Scale bar in the left panel: 20 μ m; scale bar in the right panel: 5 μ m.

cells acquire global DNA methylation levels similar to spermatozoa before birth. Compared to the number of these common peaks, we detected a relatively small number of unique peaks during spermatogonial differentiation (between THY^+ and KIT^+ spermatogonia) and during the meiosis-to-postmeiosis transition (between PS and RS) (**Fig. 2A**). In contrast, at the transition from KIT^+ spermatogonia to PS, more site-specific changes in 5mC were detected compared to the other transitions (15,285 KIT^+ unique peaks and 20,934 PS unique peaks). The distribution of these unique peaks encompasses various genomic regions, including intergenic regions, introns, exons, promotes, and upstream regions (**Fig. 2A**, right panel). A representative track view of a $\sim 100\text{kb}$ genomic region in chromosome 17 shows examples of site-specific 5mC gains and losses at the transition between KIT^+ spermatogonia and PS, and these patterns are consistent between biological replicates (**Fig. 2B**). Average tag density analysis revealed that the loss of 5mC at the KIT^+ to PS transition is pronounced at transcription start sites (TSSs) of late spermatogenesis genes, which are activated at the KIT^+ to PS transition (Sin et al., 2015) (**Fig. 2C**). Demethylation at TSSs of late spermatogenesis genes, such as *Spata16* and *Slc22a16*, was confirmed both in the MethylCap-seq data and previously-published WGBS data (Hammoud et al., 2014; Kubo et al., 2015) (**Fig. S1A**). Consistent with this finding, hypomethylation of spermatogenesis genes was recently reported in a human study (Siebert-Kuss et al., 2024). In addition, we observed extensive demethylation at the protocadherin (*Pcdh*) genes of the *Pcdh* locus (**Fig. S1B**). In contrast, 5mC levels did not change extensively in other classes of genes, such as somatic/progenitor genes that are turned off at the KIT^+ to PS transition or genes that are constitutively active or inactive during spermatogenesis (Sin et al., 2015) (**Fig. 2C**). These results suggest that DNA demethylation at TSSs is associated with activation of late spermatogenesis genes at the KIT^+ to PS transition.

DNA demethylation at TSSs correlates with the activation of late spermatogenesis genes

To further define the changes in the 5mC level at TSSs during spermatogenesis, we compared 5mC levels around all TSSs between two stages through scatter plots of MethylCap-seq read enrichment. Consistent with the peak analysis, 5mC levels around TSSs between THY^+ and KIT^+ spermatogonia and between PS and RS are highly correlated (**Fig. 3A**). KIT^+ and PS showed a decreased correlation compared to the other transitions (**Fig. 3A**), although it is still high. These data show overall moderate changes in the 5mC levels between the different cellular stages analyzed. Nevertheless, at the KIT^+ to PS transition, 909 TSSs showed a loss of 5mC levels, while 753 TSSs showed a gain of 5mC levels (**Fig. 3B**).

In mammals, approximately 70% of genes are linked to promoter CpG islands that are constitutively hypomethylated, while the rest of promoters are depleted in CpGs and subject to context-dependent DNA methylation (Weber et al., 2007). Accordingly, we found that differentially methylated TSSs at the KIT^+ to PS transition tend to overlap with low CpG promoters (**Fig. 3B**, right panel). In this context, TSSs tend to lose rather than gain 5mC levels. GO term analysis revealed that genes that lose 5mC at their TSS are enriched in genes involved in cell adhesion (presumably due to the changes in the *Pcdh* gene cluster seen earlier), sexual reproduction, spermatogenesis, and meiosis (**Fig. 3C**). This is in line with the demethylation and activation of late spermatogenesis genes described above. On the other hand, genes that show a gain of 5mC are enriched in immune-related functions. This may be related to the fact that the immune system is suppressed (and thus immune-related genes are suppressed) outside of the blood-testis barrier, where spermatogenic differentiation takes place (Cheng & Mruk, 2012).

We next examined the expression profiles of genes that lose 5mC. This group includes representative late spermatogenesis and meiotic genes, including *Spata16*, *Tdrd6*, *Slc22a16*, *Piwi1*, and *Spo11*, which are highly upregulated at the KIT^+ to PS transition (**Fig. 3D**). In contrast,

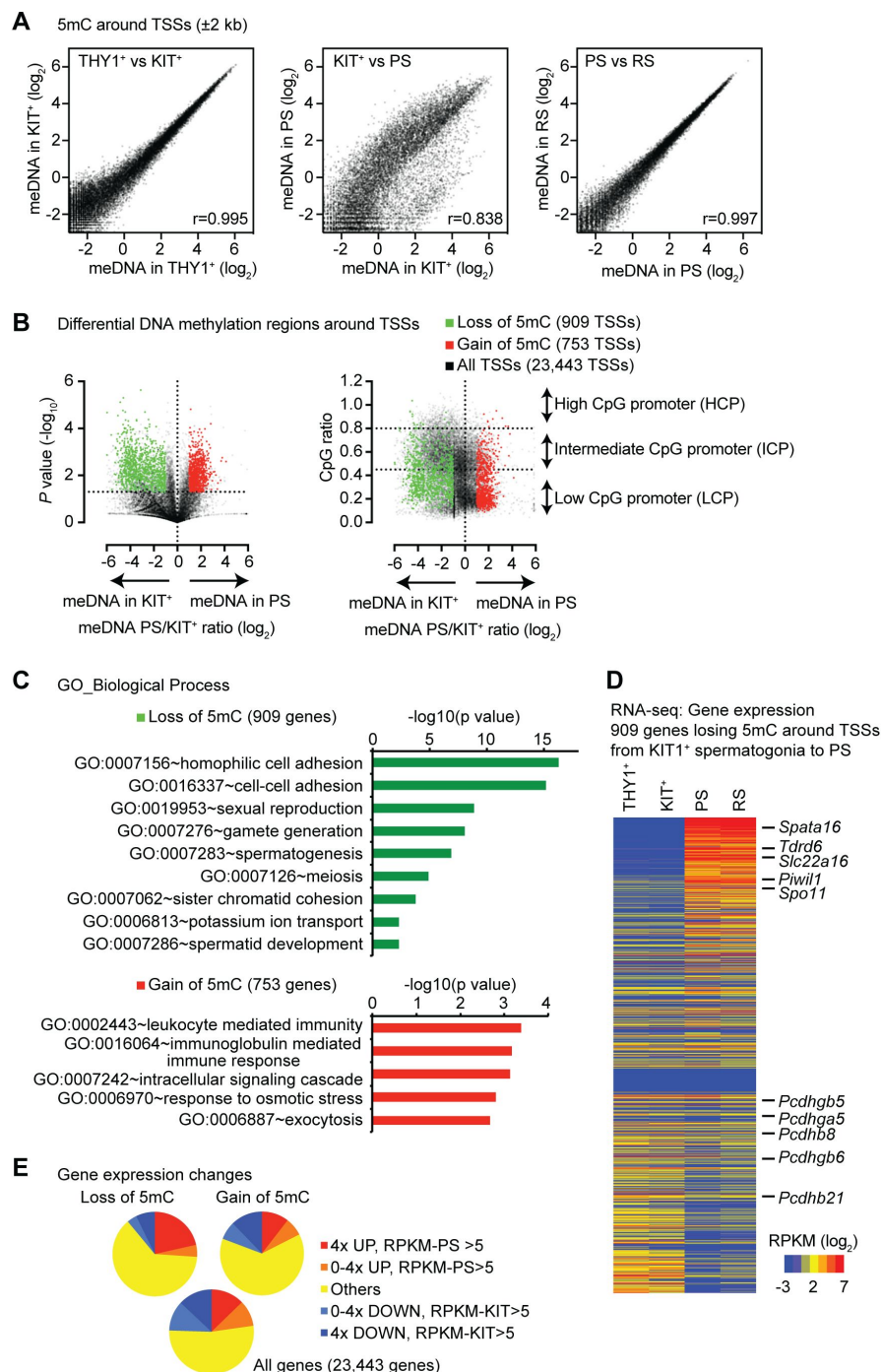


Figure 3

Changes in DNA methylation during spermatogenesis.

- (A) Enrichment analysis of MethylCap-seq around TSSs (± 2 kb) at each transition during spermatogenesis. Pearson correlation values are shown.
- (B) Volcano plots for the enrichment and P-values of distal MethylCap-seq outside ± 1 kb of TSSs during the transition from KIT⁺ spermatogonia to PS. Peaks were detected by MANorm.
- (C) GO enrichment analyses for 909 genes losing 5mC and 753 genes gaining 5mC from KIT⁺ spermatogonia to PS.
- (D) RNA-seq heatmap for 909 genes losing 5mC from KIT⁺ spermatogonia to PS.
- (E) Pie charts showing gene expression changes in three groups of genes (909 genes losing 5mC, 753 genes gaining 5mC, and all genes).

several of the *Pcdh* genes show similar expression levels during spermatogenesis (Fig. 3D). Further, genes that lose 5mC levels tend to be upregulated at the KIT⁺ to PS transition compared to genes showing a gain of 5mC levels or all genes (Fig. 3E).

One special feature of spermatogenic gene expression is a specific expression of single exon genes. Indeed, certain CpG promoters of single exon genes are hypermethylated in soma but hypomethylated in late spermatogenesis (Kato & Nozaki, 2012). We found that, among 2,848 single-exon genes, 208 lose 5mC levels at the KIT⁺ to PS transition (Fig. S2A). 50 of these genes are activated at the KIT⁺ to PS transition (Fig. S2B). Together, these data suggest that DNA demethylation at TSSs correlates with the activation of late spermatogenesis genes at the KIT⁺ to PS transition.

DNA demethylation during spermatogenesis presets the sites of nucleosome retention in spermatozoa

During spermatogenesis, histones are largely replaced with protamines. In spermatozoa, nucleosomes are retained mainly at hypomethylated TSSs, and these retained nucleosomes are implicated in paternal epigenetic inheritance (Erkek et al., 2013). Since we observed DNA demethylation at a number of TSSs at the KIT⁺ to PS transition, we hypothesized that such *de novo* hypomethylated TSSs might serve as the sites of nucleosome retention in spermatozoa. To test this hypothesis, we reanalyzed published MNase-seq data of sperm chromatin (Erkek et al., 2013). Specifically, we examined the degree of nucleosome retention (i.e., the enrichment of mononucleosomes) at various genomic regions detected in our MethylCap-seq data. Interestingly, we found a significant enrichment of mononucleosomes at promoters and exons that lost 5mC at the KIT⁺ to PS transition (termed KIT-unique methylated regions) (Fig. 4A). Consistent with these results, nucleosome retention was overserved predominantly at promoters (49%) and exons (18%) in spermatozoa (Fig. 4B).

In spermatozoa, hypomethylated promoters are enriched with histones marked by H3K4me3, which counteracts DNA methylation (Erkek et al., 2013). Thus, we next examined whether promoters and exons that lose 5mC at the KIT⁺ to PS transition acquire H3K4me3. We reanalyzed our previous H4K4me3 ChIP-seq data of KIT⁺ spermatogonia and PS (Maezawa, Hasegawa, Alavattam, et al., 2018). Our scatter plot analysis shows that at demethylated promoters and exons, there is a gain of H3K4me3 at the KIT⁺ to PS transition, and these regions tend to show mononucleosome retention in spermatozoa (Fig. 4C, 4D). These results indicate that during spermatogenesis, DNA demethylation co-occurs with a gain of H3K4-methylation and presets sites of nucleosome retention in sperm.

DNA demethylation leads to nucleosome retention and bivalent chromatin in spermatozoa

Nucleosome retention sites in sperm are implicated in paternal epigenetic inheritance (Erkek et al., 2013). Accumulating evidence suggests incorporation of the histone variant H3.3 and enrichment of bivalent marks, i.e., active H3K4me3 and silent H3K27me3, at these sites (Erkek et al., 2013; Lismer & Kimmins, 2023; Sakashita et al., 2023). Thus, we sought to characterize H3.3 incorporation and bivalent marks of the sites that lose 5mC at the KIT⁺ to PS transition in spermatozoa. To this end, we classified the MethylCap-seq peaks of KIT⁺ spermatogonia into two classes: peaks which undergo demethylation from KIT⁺ spermatogonia to PS (Class I) and all other peaks (Class II) (Fig. 5A). In spermatozoa, the Class I peaks were enriched in mononucleosomes and H3.3 compared to Class II peaks (Fig. 5A, 5B). Moreover, mononucleosome and H3.3 enrichment in spermatozoa is highly correlated at Class I peaks (Fig. 5C). These results suggest that spermatogenic demethylation precedes the incorporation of histone variant H3.3 and nucleosome retention in spermatozoa.

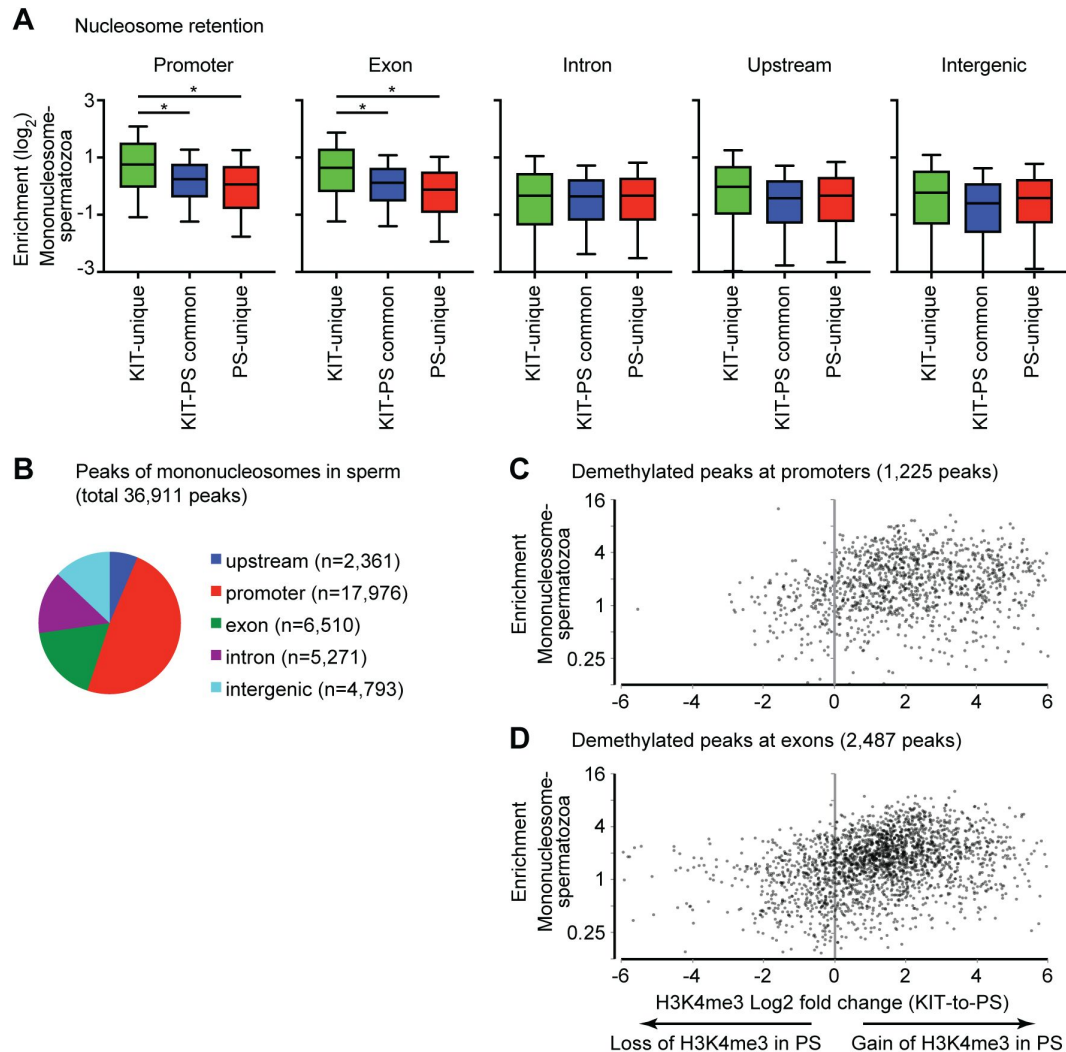


Figure 4

Sites of DNA demethylation preset sites of nucleosome retention during spermatogenesis.

(A) Box-and-whisker plots showing mononucleosome enrichment at each genomic locus for each class of genes. Central bars represent medians, the boxes encompass 50% of the data points, and the whiskers indicate 90% of the data points. * P-value < 0.0001, Mann-Whitney U test.

(B) Genomic distribution of mononucleosome peaks in sperm (total 36,911 peaks): MNase-seq data (Erkek et al., 2013). Peaks are detected by MACS2 (Padj-value<0.01, Fold enrichment>5). (C, D) Changes in the enrichment of H3K4me3 (X-axis) and mononucleosomes in spermatozoa (Y-axes) at demethylated promoters (C) and exons (D) at the mitosis-to-meiosis transition.

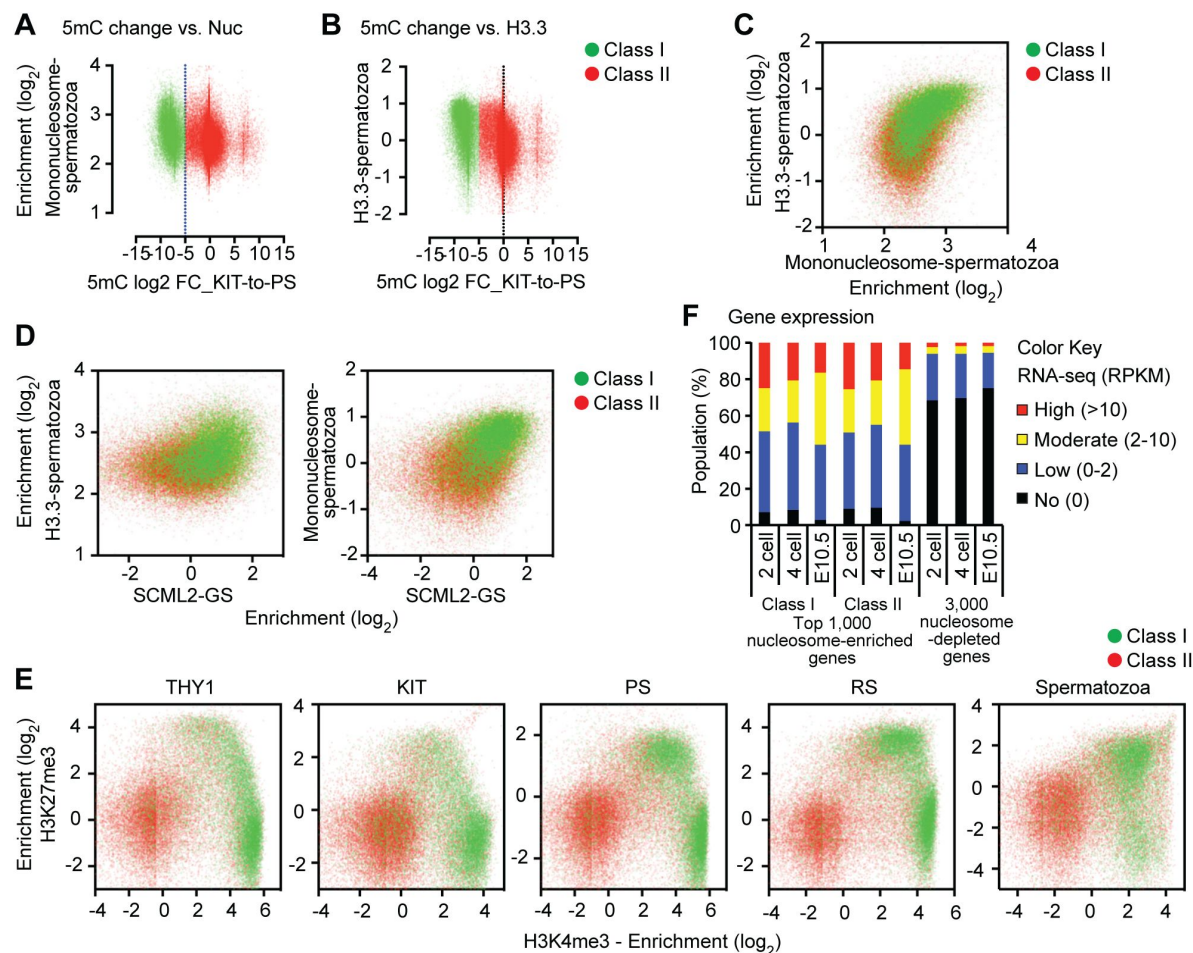


Figure 5

Sites of DNA demethylation preset sites of bivalent genomic domains in spermatozoa.

(A) Mononucleosome enrichment in spermatozoa at genomic sites of MethylCap-seq peaks of KIT⁺ spermatogonia. Class I (green): genomic sites that are demethylated at the mitosis-to-meiosis transition, and Class II (red): all other sites.

(B) H3.3 enrichment in the two classes of genomic sites in sperm.

(C) Enrichment of H3.3 and nucleosomes

(D) Enrichment of SCML2 in cultured germline stem (GS) cells, and H3.3 enrichment in sperm (left) and mononucleosome enrichment in sperm (right)

(E) Enrichment of H3K27me3 and H3K4me3 during spermatogenesis and in spermatozoa.

(F) RNA-seq analysis in embryos for three groups of genes. Top 1,000 nucleosome-enriched genes among class I and II peak-containing genes at promoters (7,577 and 7,261 genes) and bottom 3,000 nucleosome-enriched genes (Nucleosome-depleted genes).

We next wanted to examine whether bivalent marks are regulated at Class I regions during spermatogenesis. To this end, we focused on germline-specific Polycomb protein SCML2, which has a high affinity to hypomethylated DNA and is a critical regulator of germline transcriptomes and bivalent chromatin (Hasegawa et al., 2015 [↗](#); Maezawa, Hasegawa, Alavattam, et al., 2018; Maezawa et al., 2020 [↗](#); Maezawa, Yukawa, et al., 2018). SCML2 binding sites in spermatogonia predict the sites of H3K27me3 deposition in late spermatogenesis, and loss of SCML2 results in H3K27me3 depletion and defective bivalent domains (Maezawa, Hasegawa, Yukawa, et al., 2018), leading to abnormal paternal epigenetic inheritance (Sakashita et al., 2023 [↗](#)). We, therefore, examined the enrichment of SCML2 by reanalyzing our previous SCML2 ChIP-seq data (Hasegawa et al., 2015 [↗](#)). The Class I peaks tend to show a higher degree of SCML2 and H3.3 enrichment as well as a higher degree of nucleosome retention than Class II peaks (**Fig. 5D** [↗](#)). Because SCML2 binds to hypomethylated promoters enriched with H3K4me3 and establishes H3K27me3 in PS (Maezawa, Hasegawa, Alavattam, et al., 2018), we next examined enrichment of H3K4me3 and H3K27me3 at Class I and II peaks during spermatogenesis. In THY1⁺ and KIT⁺ spermatogonia, Class I peaks are enriched with H3K4me3. In contrast, Class II peaks are not enriched with H3K4me3 (**Fig. 5E** [↗](#)). The Class I peaks gradually gain H3K27me3 during the differentiation from PS to spermatozoa and the majority of Class I peaks are enriched with both H3K4me3 and H3K27me3 in spermatozoa (**Fig. 5E** [↗](#)). Taken together, we conclude that spermatogenic demethylation precedes nucleosome retention enriched with bivalent marks H3K4me3 and H3K27me3 in spermatozoa.

Finally, we investigated how spermatogenic demethylation affects gene expression in embryos. Genes whose promoters are enriched with nucleosomes in spermatozoa (top 1,000 nucleosome-enriched genes among class I and II peak-containing genes at promoters) are highly expressed in embryogenesis at 2-cell and 4-cell stages and at embryonic day 10.5 (E10.5) (**Fig. 5F** [↗](#)), although we did not observe an apparent difference between these two groups. This is highly significant when compared to genes whose promoters are depleted with nucleosomes (bottom 3000 nucleosome-enriched genes) in spermatozoa. Thus, nucleosome enrichment in spermatozoa is associated with gene expression in early embryos. These results suggest that the genes that undergo spermatogenic demethylation are enriched with nucleosomes in spermatozoa and are expressed in embryos of the next generation.

Discussion

In this study, we have mapped site-specific DNA methylation during the transition from mitotic spermatogonia to meiotic spermatocytes. We observed that promoters of a large number of late spermatogenesis genes undergo DNA demethylation at the KIT⁺ spermatogonia to PS transition, indicating that DNA demethylation might mark late spermatogenesis genes for induction. Our data also suggest that these demethylated regions predetermine sites of nucleosome retention in spermatozoa. Specifically, we propose that demethylated regions in meiotic spermatocytes acquire H3K4me3 and induce SCML2-mediated deposition of H3K27me3, thereby establishing bivalent marks at these hypomethylated nucleosome retention sites in spermatozoa (**Fig. 6** [↗](#)). In support of this model, we recently demonstrated that SCML2 mediates paternal epigenetic inheritance through sperm chromatin (Sakashita et al., 2023 [↗](#)).

Our data revealed a novel feature of the male germline in which site-specific DNA demethylation during postnatal spermatogenesis, specifically in meiosis, provides heritable information from sperm to embryos. Based on these data, we propose that meiosis is an epigenetic reprogramming process that sets up embryonic gene regulation. In 1984, Robin Holliday proposed that an aspect of the biological significance of meiosis is the reprogramming of gametes for fertilization (Holliday, 1984 [↗](#)), and our data provide a foundation for this concept.

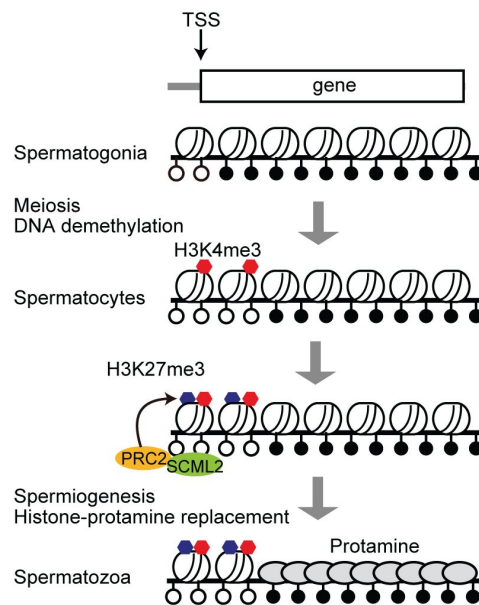


Figure 6

Model: Sites of DNA demethylation preset sites of nucleosome retention during spermatogenesis.

Genomic regions that are demethylated during the transition from mitotic spermatogonia to meiotic spermatocytes acquire H3K4me3 in meiotic spermatocytes, leading to SCML2-mediated deposition of H3K27me3, thereby establishing persisting bivalent marks at these hypomethylated nucleosome retention sites in spermatozoa.

The site-specific regulation of DNA methylation described in this study is distinct from the transient reduction of DNA methylation that occurs in the premeiotic S phase (Gaysinskaya et al., 2018 [↗](#); Huang et al., 2023 [↗](#)). This transient reduction appears to be a passive process due to delays in the establishment of maintenance DNA methylation. Therefore, it is global and not site-specific. Furthermore, the DNA remains hemimethylated. The site-specific regulation that we observe here might be an active process mediated by TET enzymes during the post-mitotic stage. In support of this possibility, TET enzymes were required in the germline (Dawlaty et al., 2013 [↗](#)). Furthermore, a recent study demonstrated that TET1 is required for the patterning of hypomethylated regions in mouse spermatozoa (Prasasya et al., 2024 [↗](#)). Alternatively, it could be caused by protection from maintenance DNA methylation and subsequent prevention of de novo DNA methylation during mitotic proliferation of spermatogonia. Further determination of the DNA demethylation mechanism in spermatogenesis is required in future studies. In addition to DNA demethylation mechanisms, a possible function of de novo DNA methyltransferases in modulating nucleosome retention in sperm was recently suggested (Fanourgakis et al., 2024 [↗](#)).

Thus, it is conceivable that DNA demethylation mechanisms and DNA methyltransferase activities are intricately regulated to shape DNA methylation profiles in spermatozoa.

The paternal genome undergoes extensive reprogramming after fertilization. For example, paternal H3.3 on promoters disappears after fertilization (Ishichi et al., 2021 [↗](#)), H3K27me3 is reprogrammed during preimplantation development (Zheng et al., 2016 [↗](#)) and the paternal genome undergoes passive and active DNA demethylation (Guo et al., 2014 [↗](#); Shen et al., 2014 [↗](#); Wang et al., 2014 [↗](#)). Thus, a major unsolved mystery is how paternal epigenetic states are inherited and maintained after fertilization. One possibility is that the paternal epigenetic states, reflecting the memory of paternal hypomethylated DNA, escape epigenetic reprogramming and persist at the promoter regions of the target genes throughout germ cell development.

Nevertheless, this possibility has been confounded by the debates about the sites of nucleosome retention in spermatozoa (Yin et al., 2023 [↗](#)). Therefore, an important next step is to elucidate how the paternal hypomethylated DNA regions are regulated in the sperm-to-zygote transition.

Materials and methods

Animals

Mice on the C57BL/6 background were maintained and used according to the guidelines of the Institutional Animal Care and Use Committee (protocol no. IACUC2018–0040) at Cincinnati Children’s Hospital Medical Center.

Germ cell fractionation

Wild-type mice on the C57BL/6 background (at least 12 independent mice at 8–12 weeks of age or at least 20 independent mice at 6–8 days of age) were used for isolation of germ cells.

Pachytene spermatocytes and round spermatids were isolated via BSA gravity sedimentation as previously described (Bellve, 1993 [↗](#)). Purity was confirmed by nuclear staining with Hoechst 33342 using fluorescence microscopy. In keeping with previous studies from the Namekawa lab (Alavattam et al., 2019 [↗](#); Hasegawa et al., 2015 [↗](#); Maezawa, Hasegawa, Yukawa, et al., 2018; Maezawa, Yukawa, et al., 2018), only fractions with a mean purity of ≥90% were used to extract genomic DNA.

Spermatogonia were isolated as described previously (Hammoud et al., 2014 [↗](#)) and collected from C57BL/6 wild-type mice aged 6–8 days. Testes were collected in a 24-well plate in Dulbecco’s Modified Eagle Medium (DMEM) supplemented with GlutaMax (Thermo Fisher Scientific), non-

essential amino acids (NEAA) (Thermo Fisher Scientific), and penicillin and streptomycin (Thermo Fisher Scientific). After removing the *tunica albuginea* membrane, testes were digested with collagenase (1 mg/ml) at 34°C for 20 min to remove interstitial cells, then centrifuged at 188×g for 5 min. Tubules were washed with the medium and then digested with trypsin (2.5 mg/ml) at 34°C for 20 min to obtain a single-cell suspension. Cells were filtered with a 40-μm strainer to remove Sertoli cells, and the cell suspension was plated in a 24-well plate for 1 h in the medium supplemented with 10% fetal bovine serum, which promotes adhesion of remaining somatic cells. Cells were washed with magnetic cell-sorting (MACS) buffer (PBS supplemented with 0.5% BSA and 5 mM EDTA) and incubated with CD117 (KIT) MicroBeads (Miltenyi Biotec) on ice for 20 min. Cells were washed and resuspended with MACS buffer and filtered with a 40-μm strainer. Cells were separated by autoMACS Pro Separator (Miltenyi Biotec) with the program “possel.” Cells in the flow-through fraction were washed with MACS buffer and incubated with CD90.2 (THY1) MicroBeads (Miltenyi Biotec) on ice for 20 min. Cells were washed and resuspended with MACS buffer and filtered with a 40-μm strainer. Cells were separated by autoMACS Pro Separator (Miltenyi Biotec) with the program “possel.” Purity was confirmed by immunostaining.

Quantification of 5mC

Genomic DNA was purified from isolated germ cells using Genomic DNA Clean & Concentrator Kit (ZymoResearch) according to the manual provided. The percentage of 5mC in total DNA was quantified by MethylFlash™ Methylated 5mC DNA Quantification Kit (Colorimetric) (Epigentek) according to the manual provided.

Histological analysis and immunofluorescence

For the preparation of testicular paraffin blocks, testes were fixed with 4% paraformaldehyde (PFA) overnight at 4°C with gentle inverting. Testes were dehydrated and embedded in paraffin. For immunofluorescence analysis of testicular sections, antigen retrieval was performed by boiling the slides in Target Retrieval Solution (DAKO) for 10 min and letting the solution cool for 30 min. For the detection of 5mC, sections were incubated in 4 N HCl for 1 h at room temperature and then neutralized in 100 mM Tris-HCl (pH 8.5) for 10 min, followed by a standard immunostaining protocol. Sections were blocked with Blocking One Histo (Nacalai) for 1 h at room temperature and then incubated with anti-5mC (1/200 dilution, Eurogentec, clone 33D3), anti-ZBTB16 (1/200 dilution, Santa Cruz, sc22839), and anti-H1T (1/1000 dilution, gift from Dr. Mary Ann Handel) antibodies overnight at 4°C. The resulting signals were detected by incubation with secondary antibodies conjugated to Alexa Fluor dyes (Thermo Fisher Scientific or Jackson ImmunoResearch). Sections were counterstained with DAPI. Images were obtained via a laser scanning confocal microscope A1R (Nikon) and processed with NIS-Elements (Nikon) and ImageJ (National Institutes of Health) (Schneider et al., 2012 [DOI](#)).

MethylCap-seq

Genomic DNA was fragmented by sonication and methylated DNA fragments were captured by MethylAffinity Methylated DNA Enrichment Kit (GeneCopoeia). DNA libraries were created by ChIPmentation method (Schmidl et al., 2015 [DOI](#)). Briefly, methylated DNA fragment-bound beads were resuspended in 30 μl of the tagmentation reaction buffer (10 mM Tris-HCl pH 8.0 and 5 mM MgCl₂) containing 1 μl Tagment DNA Enzyme from the Nextera DNA Sample Prep Kit (Illumina) and incubated at 37°C for 10 min in a thermal cycler. The beads were washed twice with 150 μl cold wash buffer (50 mM Tris-HCl pH 8.0, 150 mM NaCl, 1 mM EDTA, 0.1% SDS, 0.1% NaDOC, and 1% Triton X-100), incubated with elution buffer (10 mM Tris-HCl pH 8.0, 1 mM EDTA, 250 mM NaCl, 0.3% SDS, 0.1 μg/μl Proteinase K) at 42°C for 30 min. DNA was purified with the MinElute Reaction Cleanup Kit (Qiagen) and amplified with NEBNext High-Fidelity 2× PCR Master Mix (NEB). Amplified DNA was purified by Agencourt AMPure XP (Beckman Coulter). Afterward, DNA

fragments in the 250-to 500-bp size range were prepared by agarose gel size selection. DNA libraries were adjusted to 5 nM in 10 mM Tris-HCl pH 8.0 and sequenced with an Illumina HiSeq 2500.

Data analysis

Data analysis for MethylCap-seq, ChIP-seq, and RNA-seq was performed in the BioWardrobe Experiment Management System (Kartashov & Barski, 2015 [DOI](#)). For MethylCap-seq and ChIP-seq analysis, reads were aligned to mouse genome mm10 with Bowtie (version 1.2.0, (Langmead et al., 2009 [DOI](#))), assigned to NCBI RefSeq genes or isoforms, and coverage was displayed on a local mirror of the UCSC genome browser (Meyer et al., 2013 [DOI](#)). MethylCap-seq peaks were identified using MACS2 (version 2.1.1.20160309, (Zhang et al., 2008 [DOI](#))) with a default parameter setting for narrow peak detection in BioWardrobe. Pearson correlations for the genome-wide enrichment of the peaks among MethylCap-seq library replicates were analyzed using SeqMonk (Babraham Institute).

MANorm, software designed for quantitative comparisons of ChIP-seq data sets (Shao et al., 2012 [DOI](#)), was used to compare the genome-wide MethylCap-seq peaks among stages in spermatogenesis. Unique peaks were defined using the following criteria: (1) defined as “unique” by the MANorm algorithm; (2) P-value <0.01; (3) raw counts of unique reads >10. Peaks common to two stages were defined using the following criteria: (1) defined as “common” by MANorm algorithm; (2) raw read counts of both stages >10.

Average tag density profiles were calculated around transcription start sites for gene sets of somatic/progenitor genes, late spermatogenesis genes, constitutive active genes, and constitutive inactive genes, as described previously (Sin et al., 2015 [DOI](#)). The resulting graphs were smoothed in 200-bp windows. Enrichment levels for MethylCap-seq experiments were calculated for 10-kb windows, promoter regions of genes (± 5 kb surrounding TSSs).

Enrichment levels for MethylCap-seq and ChIP-seq experiments were calculated for 4-kb windows, promoter regions of genes (± 2 kb surrounding TSSs). To normalize tag values, read counts were multiplied by 1,000,000 and then divided by the total number of reads in each nucleotide position. The total amount of tag values in promoter or enhancer regions was calculated as enrichment.

To detect differentially methylated regions around TSSs among stages in spermatogenesis, a read count output file was input to the DESeq2 package (version 1.16.1); then, the program functions DESeqDataSetFromMatrix and DESeq were used to compare each gene's expression level between two biological samples. Differentially methylated regions were identified through binominal tests, thresholding Benjamini-Hochberg-adjusted P values to <0.05. To perform gene ontology analyses, the functional annotation clustering tool in DAVID (version 6.8) was used, and a background of all mouse genes was applied.

For RNA-seq analysis, reads were aligned by STAR (version STAR_2.5.3a) (Dobin et al., 2013 [DOI](#)) (Dobin et al., 2013 [DOI](#)) with default arguments except `—outFilterMultimapNmax 1` and `—outFilterMismatchNmax 2`. The `—outFilterMultimapNmax` parameter was used to allow unique alignments only, and the `—outFilterMismatchNmax` parameter was used to allow a maximum of 2 errors. NCBI RefSeq annotation from the mm10 UCSC genome browser was used, and canonical TSSs (1 TSS per gene) were analyzed. All reads from the resulting bam files were split for related isoforms with respect to RefSeq annotation.

Data availability

NGS datasets used in this study are publicly available and referenced within the article. All the MethylCap-seq data generated in this study are deposited to the Gene Expression Omnibus (GEO) under accession code GSE262669.

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Author Contributions

S.M. and S.H.N. designed research; S.M. performed research; S.M., M. Y., A.S. A.B., and S.H.N. analyzed data; and S.M. and S.H.N. wrote the paper. S.H.N. supervised the project.

Additional files

Supplemental Figures [↗](#)

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Reviewer #1 (Public review):

This study investigates the role of site-specific DNA methylation changes during spermatogenesis and their contribution to paternal epigenetic inheritance. Using MethylCap-seq, the authors identify a transient, site-specific loss of DNA methylation at transcription start sites (TSSs) of late spermatogenesis genes during the transition from differentiating spermatogonia (KIT+) to pachytene spermatocytes (PS). This demethylation event correlates with the gain of H3K4me3, which presets nucleosome retention sites in mouse sperm. The study proposes that selective loss of DNA methylation at a subset of promoters is required for nucleosome retention and the establishment of epigenetic states that may influence embryonic gene regulation. These findings provide complementary insights to earlier work by the Peters lab, "DNA methylation modulates nucleosome retention in sperm and H3K4 methylation deposition in early mouse embryos."

Overall, the study presents a valuable dataset; however, additional analyses could strengthen the conclusions and provide further mechanistic insights.

Major Comments:

(1) Prior work should be acknowledged and used for comparative analysis. A key proposal in this study is that regions undergoing DNA methylation loss retain histones, influencing the zygote's epigenetic landscape. However, previous studies (e.g., Peters et al.) have shown that regions losing methylation in DNMT3a/b knockout (KO) mice do not necessarily retain histones, suggesting additional factors are involved. Moreover, Peters et al. demonstrated that regions of low DNA methylation in sperm render paternal alleles permissive for H3K4me3 establishment in early embryos, independent of the paternal inheritance of sperm-borne H3K4me3. Comparing these findings would refine the model presented in this study.

(2) Figure 2A: The data suggest an increase in methylation peaks in PS cells. How does this align with the hypomethylation observed in Figure 1D? Reconciling these observations would improve clarity.

(3) Figure 4A: The effect size of demethylation on nucleosome retention is unclear - do all demethylated promoters retain histones or only a subset? Quantifying this would clarify whether DNA methylation loss consistently predicts nucleosome retention.

(4) Prior studies have generated bisulfite sequencing data from Tet KO sperm. Do the regions that undergo demethylation during the KIT⁺ to PS transition overlap with those misregulated in TET KO sperm? Integrating this comparison could provide further insight into the regulation of site-specific demethylation.

(5) The role of SCML2 enrichment in germline stem cells and its connection to H3K27me3 deposition in later germ cells is unclear. Earlier figures show that regions undergoing DNA demethylation from KIT⁺ to PS include genes expressed in later-stage germ cells.

Why is SCML2 enrichment occurring in germline stem cells (GSCs)? Why is H3K27me3 only acquired at later stages if SCML2 is already present? Is SCML2 preventing premature expression independent of K27ME?

Showing the dynamics of H3K27me3 and SCML2 across these stages would clarify the proposed conclusions.

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Reviewer #2 (Public review):

Summary:

This study profiles the genome-wide distribution of DNA methylation using methylation capture sequencing in four stages of male germ cells: Thy1⁺ (undifferentiated spermatogonia), Kit⁺ (differentiated spermatogonia), pachytene spermatocytes, and round spermatids. These analyses revealed site-specific loss of DNA methylation in pachytene cells compared with differentiating spermatogonia. Integrated analysis using published datasets indicates that hypomethylated sites correlate with nucleosome retention sites and bivalent histone methylation sites in sperm.

Strengths:

The methyl-seq approach provides a comprehensive profile of DNA methylation in male germ cells. The concept that DNA hypomethylation in meiotic cells precedes histone modification and histone retention in sperm is interesting.

Weaknesses:

(1) In the title, the word "presets" should be changed to "precedes" or "correlates with". Preset means a causal relationship, which is not the case. This needs to be changed throughout the manuscript. For example, in the abstract, "predetermine" needs to be changed to "precede".

(2) The statement that "Based on these results, we propose that meiosis is a process of epigenetic reprogramming that sets up embryonic gene regulation" (lines 94-95) is a speculation that in the opinion of this reviewer should be removed from the text. It is too broad and not supported by the data presented.

(3) Figure 1B: details are missing. How many cells were analyzed/used? How many times was this experiment done [(The number of experiments (n))]? Were the changes statistically significant (Lines 109-111)?

(4) Figure 1A and Figure 1D: These seem to be contradictory. According to Figure 1D, leptotene/zygotene spermatocytes show bright 5mC staining. However, the diagram in 1A shows delayed recovery of DNA methylation. The authors should clarify this. It appears that 5mC was high in Kit⁺ spermatogonia and leptotene/zygotene spermatocytes, and then decreased in pachytene spermatocytes.

(5) L121-122: Statement: These results suggest that 5mC levels change dynamically during spermatogenesis before and after the transient reduction of DNA methylation in the premeiotic S phase. In order to make this claim about the premeiotic S phase, I suggest performing 5mC staining in premeiotic S phase cells, which can be pulse-labelled with BrdU or cite a reference if available.

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